

TEACHER IMPLEMENTATION KIT

Ms. Lang's VIQRC

Competitive Elective

Six-person teams | VEX IQ Level Up | 2026-27

IMPLEMENTATION ITEM	CURRENT SETTING
Course pattern	Short Monday instruction; long Wednesday STEM Lab; long Friday Level Up lab
Team model	Six students: two drivers, two programmers, two strategists; lead and backup in each pair
Evidence model	One shared engineering notebook per team plus individual Canvas evidence
Curriculum site	https://spartandesign.github.io/spartandesign-viqrc/

Outcome Use this guide to move from files to a functioning course: import Canvas, personalize local details, form teams, establish notebook routines, publish the first modules, and maintain the course through every grading period.

30-minute quick launch

- Open the editable family syllabus and handbook. Replace placeholder contact, room, period, event, travel, and local-policy language.
- Create a blank Canvas sandbox and import 01_CANVAS/stauffer-viqrc-canvas.imsc. Leave everything unpublished.
- Set the Canvas course dates to August 13, 2026 through June 4, 2027, then confirm the district time zone and grading periods.
- Publish only the Start Here module after testing it in Student View. Add due dates manually from the six grading windows in this guide.
- Duplicate the engineering-notebook PowerPoint once per team and preserve the original as a clean master.
- Share the public curriculum site with students and families; use Canvas for submissions, grades, announcements, and local details.

1. Know what each file does

FOLDER	FILE	HOW MS. LANG USES IT
00_START_HERE	Implementation guide PDF / DOCX	Follow the launch sequence; edit the Word file if local procedures change.
00_START_HERE	Level Up v1.1 curriculum audit	See which manual rules drive site, Canvas, handbook, and notebook content.
01_CANVAS	stauffer-viqrc-canvas.imsc	Import into a blank Canvas course, add dates, then publish in stages.
02_ENGINEERING_NOTEBOOK	Six-person team notebook PPTX	Make one working copy per team; students duplicate the entry slides.
03_FAMILY_AND_STUDENT	Family syllabus PDF / DOCX	Send or print the short family-facing course overview; edit the DOCX first.
03_FAMILY_AND_STUDENT	VIQRC handbook PDF / DOCX	Use for detailed procedures, competition expectations, and acknowledgment.
04_TEAM_SETUP	team-roster-template.csv	Assign role pairs and distribute returning members across teams.
05_LINKS	Curriculum_Site_and_Official_Links.md	Open the public curriculum and current official resources.

Source boundary The Level Up manual itself is not bundled. Always use the live official manual because current rules and updates supersede every summary in this kit.

Personalize before students see the course

- Confirm course title, grades served, room, period, teacher contact method, office hours, and school links.
- Confirm course dates, six grading-period due dates, late-work and missing-work policies, and any eligibility requirements.
- Replace generic event language with confirmed tournaments, permission deadlines, transportation, travel, and family volunteer information.
- Insert district-approved language for accessibility, academic integrity, media permission, fundraising, safety, and student conduct.
- Decide whether families sign only the short syllabus or the full handbook acknowledgment page.
- Confirm the number of six-person teams and how the six or seven returning students will be distributed.

Keep these distinctions intact

- The competition pathway uses VEX Robotics and the Global Robotics & Science Foundation, not RECF.
- Classroom role pairs are drivers, programmers, and strategists. The event Drive Team is separate: Driver 1, Driver 2, and Loader.
- Students are graded on evidence, process, preparation, and growth - never on trophies, rankings, or a partner team's performance.

2. Import and configure Canvas

Import into a blank sandbox

- Open the blank Canvas course and select Settings, then Import Course Content.
- Choose Canvas Course Export Package. If the district labels .imscs imports as Common Cartridge 1.x Package, choose that equivalent option.
- Upload stauffer-viqrc-canvas.imscs and select All Content. Do not import into a live course that already contains similarly named content.
- Wait for the import to show Completed. Review any issue log before opening the imported content.
- Leave the course and every imported module unpublished until the configuration checklist is complete.

Important The shell uses suggested checkpoint language rather than authoritative due-date timestamps. Set dates manually from the confirmed school grading calendar after import.

Post-import configuration checklist

- Settings: course name, course code, term, time zone, participation dates, navigation, and license visibility.
- Grades: confirm category weights of 30%, 25%, 20%, 15%, and 10%; then set late and missing policies.
- Assignments: set the six checkpoint dates, verify point values, and revise any activity Ms. Lang will not grade.
- Modules: keep later modules unpublished; verify prerequisites and the order of pages, discussions, assignments, and rubrics.
- Home page: use Modules or the imported welcome page, then remove unused navigation links.
- Links: test the curriculum site, current manual, Global Robotics resources, STEM Labs, and school links.
- Groups: after team selection, create a VIQRC Teams group set with six students per Canvas group.
- Calendar: add confirmed events, travel and permission deadlines, notebook checks, mock events, and showcase dates.
- Student View: complete the Start Here path and test every download and submission type.

Recommended publication sequence

WHEN	PUBLISH	MS. LANG'S ACTION
Before class	Start Here	Add local policies, contact details, syllabus, welcome announcement, and baseline tasks.
Week 1	Competition Foundations	Teach field/scoring vocabulary, manual navigation, rule tags, safety, and event roles.
Each grading period	One active unit	Publish the unit after its due dates, resources, and current-rule references are checked.
As confirmed	Event materials	Publish travel, permission, packing, inspection, and family information only when finalized.

3. Launch the first full week

The course pattern is a short Monday followed by longer Wednesday and Friday blocks. Use the first full week to establish the rhythm students will experience all year.

DAY	LESSON PURPOSE	STUDENT WORK	EVIDENCE BEFORE EXIT
Monday	Course promise, safety, public site, Canvas navigation, and six-person role pairs	Baseline survey; syllabus; lab entry and cleanup practice	Individual baseline plus one safety commitment
Wednesday	VEX IQ system orientation and STEM Lab routine	Inventory components; complete a selected STEM Lab build/coding task	Image or screenshot, observation data, and one technical question
Friday	Level Up field, scoring, manual navigation, and evidence routine	Rules jigsaw; manual citations; simple scoring challenge	Notebook entry: target, evidence, decision, and next test

First-week teacher setup

- Prepare one labeled VEX IQ kit and one parts-inventory workspace per prospective team.
- Choose the first Wednesday STEM Lab and confirm required materials and devices.
- Open the current Level Up manual and official field/scoring resources before Friday; do not teach from an old printed summary alone.
- Collect baseline information on experience, preferred roles, collaboration, coding, driving, and attendance constraints.
- Use returning students as distributed mentors, not permanent owners of the robot, code, notebook, or decisions.

4. Form and manage six-person teams

PAIR	LEAD RESPONSIBILITY	BACKUP RESPONSIBILITY	SHARED EVIDENCE
Drivers	Own practice plans, repeatable routes, and match communication	Reproduce routes, identify issues, and step in	Timed runs, error patterns, match notes, and handoff plans
Programmers	Own code architecture, version labels, and autonomous tests	Reproduce deployments, debug, and maintain continuity	Code captures, test tables, failures, and version changes
Strategists	Own rules, scoring models, scouting, and test priorities	Verify citations, challenge assumptions, and maintain contingencies	Rule tags, scoring math, decision matrices, and partner plans

Balanced-team procedure

- Review baseline evidence, preferences, support needs, attendance patterns, and collaboration strengths.
- Distribute the six or seven returning members so expertise is not concentrated in one team.
- Create six-person rosters with two students per role pair. Name a lead and backup, but hold both accountable for shared evidence.
- Enter rosters in team-roster-template.csv and create matching Canvas groups.
- Run a two-week role trial; revise placements based on participation and evidence, not status or prior experience alone.
- Rotate practice experiences so every student develops basic building, coding, driving, strategy, and notebook fluency.

Event-role clarification

Drive Team For official matches, identify Driver 1, Driver 2, and Loader according to the current Level Up manual. Classroom 'drivers' do not automatically equal the event Drive Team.

Weekly role check

- Each pair reports one completed task, one piece of evidence, and one next action.
- Lead and backup can each explain the current mechanism, code, or strategy decision.
- If work is invisible in the notebook or Canvas, the team treats it as unfinished.
- Ms. Lang intervenes when one student controls access or a backup cannot reproduce the work.

5. Implement the engineering notebook

Create one working notebook per team

- Keep the original PowerPoint as a clean master that students cannot overwrite.
- Duplicate it once per team with a consistent name such as Team_12345A_Level_Up_Engineering_Notebook.pptx.
- Give all six students edit access and give Ms. Lang owner or manager access.
- Complete opening slides once: identity, roles, norms, Level Up overview, manual tracker, design-process tutorial, and calendar.
- Duplicate entry slides for every meaningful test, build change, code revision, strategy change, event, and reflection.
- At each grading window, submit the shared notebook plus individual reflection or evidence in Canvas.

What a complete entry contains

ENTRY	WHAT STUDENTS RECORD	ACCEPTABLE EVIDENCE
Target	A measurable problem, goal, or rule-based constraint	Score, time, accuracy, failure limit, or inspection requirement
Plan	What changes and what stays controlled	Sketch, pseudocode, mechanism photo, matrix, or procedure
Test	Enough detail for another student to repeat it	Trials, setup, measurement, manual version, and rule tags
Evidence	Observed results, including failures	Data table, annotated photo, code capture, match log, or video link
Decision	Keep, revise, replace, or retest - with reasoning	Claim tied to evidence and constraints
Next step	Owner, date, and next controlled test	Lead/backup handoff and the next test question

Daily closeout routine

Reserve the final 8-12 minutes of every long block for documentation and reset. One student may type, but all six contribute evidence or reasoning.

- Capture the robot, code, setup, and data before anything is disassembled or overwritten.
- Record target, change, controlled variables, trials, results, decision, and next owner.
- Add manual version and rule tags when a rule affects design, code, strategy, inspection, or event decisions.
- Complete inventory, battery, field, device, and floor cleanup before dismissal.

Teacher review Ask: Can another student reproduce the work? Does evidence support the decision? Can lead and backup explain it? What must be tested next?

6. Teach the design process as a visible cycle

Students use the design process every week, not only at a project's end. The cycle may repeat in minutes during a test or across weeks during a redesign.

STEP	STUDENT QUESTION	NOTEBOOK PROOF
1. Define	What measurable problem and constraints matter?	Problem, baseline, rule tags, manual version, and criteria
2. Research	What do the manual, STEM Lab, data, and failures tell us?	Source notes, observations, prior trials, and research
3. Imagine	What are at least two plausible approaches?	Sketches, flowcharts, mechanisms, routes, or scoring models
4. Select	Why is this idea the best current choice?	Decision matrix or tradeoff explanation
5. Build/code	What exactly changed from the previous version?	Version label, annotated photo, notes, or code capture
6. Test	Can someone repeat the measurable test?	Procedure, variables, trials, raw results, and failures
7. Decide	Does evidence support keep, revise, replace, or retest?	Claim, evidence, reasoning, and next test
8. Communicate	Can the team explain and transfer the work?	Design review, handoff, reflection, or portfolio artifact

Evidence sentence frame

Claim + evidence + reasoning We decided to [keep/revise/replace/retest] _____ because [trial data] _____. This matters because [rule, constraint, or goal] _____. Next, [student or pair] will test _____ by [date].

Prevent common notebook failures

Use these failure patterns as a short Monday critique before teams plan their next test.

- Photo dump: require a caption explaining what changed and why the image matters.
- Success-only history: grade useful failure evidence and the quality of the next decision.
- After-the-fact writing: close each long block with documentation before cleanup ends.
- One-person notebook: require pair contributions and individual Canvas evidence at checkpoints.
- Untraceable rules: require manual version and rule tags.
- Unrepeatable tests: require setup, variables, trials, units, results, and next question.

Fast correction Project one anonymous notebook entry. Students identify one missing evidence element, write the revision they would make, and then repair their team's latest entry before beginning new work.

7. Run weekly and grading-period routines

Weekly operating rhythm

DAY	PRIMARY MODE	TEACHER MOVE	REQUIRED EVIDENCE
Monday	Direct instruction or demo	Model one rule, mechanism, code pattern, strategy method, or notebook move	Individual check plus team test target
Wednesday	VEX IQ STEM Lab	Coach structured building, coding, sensing, driving, or data practice	Result, image or code capture, data, and reflection
Friday	Level Up competition lab	Coach scoring, mechanisms, autonomous, driving, strategy, judging, inspection, or mock matches	Test entry with version, rule tags, result, decision, and next step

Grade evidence, not outcomes

CATEGORY	WEIGHT	PRIMARY EVIDENCE
Engineering process	30%	Notebook entries, sketches, prototypes, tests, decisions, and iterations
Technical performance	25%	Build, code, driving reliability, measurable skills growth, and troubleshooting
Collaboration	20%	Role follow-through, lead/backup continuity, communication, safety, stewardship
Competition readiness	15%	Rules, inspection, strategy, scouting, judging, preparation, and conduct
Reflection + communication	10%	Reviews, portfolios, demonstrations, reflections, and handoffs

Six notebook checkpoints

WINDOW	CHECKPOINT	MINIMUM TEAM EVIDENCE
Sep 10-18	Progress I	Safety; charter; rules jigsaw; baseline build, drive, and score data
Oct 22-30	Progress II	Mechanism alternatives; matrix; controlled tests; legal robot v1
Dec 10-18	Semester I	Driver route; autonomous; mock event/interview; portfolio defense
Feb 4-12	Progress IV	Failure analysis; robot v2 proposal; A/B tests; benchmark
Apr 8-16	Progress V	Partner playbook; readiness; judge practice; event reflection
May 27-Jun 4	Semester II	Final portfolio; showcase; clinics; next-season handoff

At every checkpoint, score the shared notebook and require a short individual Canvas reflection so quiet work and accountability remain visible.

8. Maintain competition readiness

Before adding an event to Canvas

- Confirm the event in the official VEX events system: program, grade level, location, date, status, and organizer.
- Confirm school approval, transportation, travel, supervision, permission, cost, meals, arrival, dismissal, and family communication.
- Publish one event page with schedule, contacts, packing, inspection, student jobs, deadlines, and contingency plan.
- Use current official resources for field, scoring, robot, Drive Team, inspection, judging, and conduct rules.

Event readiness gate

AREA	READY WHEN	EVIDENCE
Robot	Legal, labeled, charged, secure, and reproducible after transport	Inspection, inventory, spares, photos, and repair plan
Code	Correct program is backed up, versioned, deployable, and understood	Version log, deployment test, and autonomous results
Drive Team	Driver 1, Driver 2, and Loader understand current responsibilities	Match simulations, communication plan, and contingencies
Strategy	Team explains priorities, partner communication, and legal contingencies	Scouting, match plan, citations, and scoring model
Notebook/judging	Team finds evidence quickly and explains authentic work	Notebook index, practice questions, and peer feedback
Logistics	Permissions, travel, meals, contacts, schedule, attire, and equipment confirmed	Completed checklist and family communication

Rule-update protocol

- Check the live manual and official resources before each unit, before every event, and on scheduled update dates.
- Record the new version and publication date in every team's manual tracker.
- Identify changed rule tags and affected design, code, strategy, inspection, Drive Team, judging, or event materials.
- Teach the change in the next Monday mini-lesson and require teams to revise assumptions.
- Retest anything affected and record evidence before the next event.
- Update Canvas and the public curriculum only after verification against the official source.

Manual tracker: This kit is aligned to Level Up Game Manual Version 1.1, released August 6, 2026. A later official version supersedes this guide, Canvas, site, handbook, and notebook prompts.

9. Launch families and the first 30 days

Family launch

- Edit the family syllabus DOCX, verify local details, then export a new PDF before distribution.
- Decide whether the handbook is required reading or a reference; clearly state what families acknowledge.
- Share the public site and explain Canvas holds announcements, grades, due dates, submissions, and event details.
- Explain that events may require additional permission, travel, transportation, time, cost, or eligibility information.
- Give families one reliable contact method and one place to find the current schedule.

First 30 days

TIME	IMPLEMENTATION TARGET	MS. LANG VERIFIES
Before Day 1	Canvas Start Here, policies, syllabus, devices, kits, safety, and site link are ready	Student View works; later modules remain unpublished
Week 1	Students understand rhythm, manual navigation, evidence, and baseline expectations	Every student submits baseline evidence
Week 2	Balanced six-person teams begin a two-week role trial	Roster and Canvas groups match; returners are distributed
Week 3	Teams maintain notebooks and complete repeatable STEM Lab and Level Up tests	Lead and backup can reproduce the work
Week 4	Teams defend evidence-based decisions and prepare for Progress I	Entries contain targets, tests, evidence, decisions, next steps, and citations

Implementation is complete when

- Families have the verified syllabus, correct link, and a reliable contact path.
- Six-person teams and Canvas groups match; role leads and backups are visible.
- Every team has a shared notebook and uses daily closeout.
- Monday, Wednesday, and Friday tasks produce expected evidence.
- Six checkpoints and category weights are represented in Canvas.
- Confirmed events have local logistics and current official-rule references.

10. Official links and ongoing maintenance

RESOURCE	URL	USE
Public curriculum	https://spartandesign.github.io/spartandesign-viqr/	Student/family pathway, rhythm, grading, milestones, downloads
Current manual	https://link.vex.com/docs/26-27/viqr/game-manual	Authoritative rules; verify version before teaching or competing
Global Robotics	https://www.globalrobotics.org/	Competition pathway and foundation information
VIQRC team resources	https://library.globalrobotics.org/hc/en-us/categories/51091266492436-Teams?sc=vex-iq	Official Level Up and team resources
VEX events	https://events.vex.com/	Official event discovery and registration information
VEX IQ STEM Labs	https://education.vex.com/stemlabs/iq/teacher-resources	Wednesday activities and teacher resources
Canvas import help	https://community.instructure.com/en/kb/articles/660728-unknown	Official Canvas package-import procedure
Canvas date adjustment	https://community.instructure.com/en/kb/articles/660737-how-do-i-adjust-events-and-due-dates-in-a-course-import	Official import-date guidance

Monthly maintenance

- Test external links and replace any retired or redirected official resource.
- Review the live manual version and update trackers, Canvas, site, and printed language when required.
- Compare Canvas dates with the school calendar, event calendar, and grading windows.
- Archive strong anonymous examples of design evidence, code testing, strategy, interviews, and handoffs.
- Use feedback and checkpoint data to adjust pacing while preserving evidence and safety expectations.

One source of truth Use the public site for the stable pathway, Canvas for classroom operations, the notebook for team engineering evidence, and the live official manual for competition rules.